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LARGE LANGUAGE MODELS HAVE INTRINSIC SELF-CORRECTION ABILITY

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ABSTRACT

Large language models (LLMs) have attracted significant attention for their exceptional abilities in various natural language processing tasks, but they suffer from hallucinations that will cause performance degradation. One promising solution to improve the LLMs' performance is to ask LLMs to revise their answer after generation, a technique known as self-correction. Among the two types of self-correction, intrinsic self-correction is considered a promising direction because it does not utilize external knowledge. However, recent works doubt the validity of LLM's ability to conduct intrinsic self-correction. In this paper, we present a novel perspective on the intrinsic self-correction capabilities of LLMs through theoretical analyses and empirical experiments. In addition, we identify two critical factors for successful self-correction: zero temperature and fair prompts. Leveraging these factors, we demonstrate that intrinsic self-correction ability is exhibited across multiple existing LLMs. Our findings offer insights into the fundamental theories underlying the self-correction behavior of LLMs and remark on the importance of unbiased prompts and zero temperature settings in harnessing their full potential.

1 INTRODUCTION

Large Language Models (LLMs) have transformed the field of natural language processing, showcasing exceptional capabilities in generating coherent, contextually appropriate text across diverse domains (OpenAI, 2024; Gao et al., 2024). As of now, while LLMs are still not considered equivalent to human-level intelligence (Shen et al., 2023; Mathur et al., 2024), they exhibit some interesting characteristics that are possessed by human beings, including the emergent ability (Wei et al., 2022a)[1] and performing complex tasks (Chiang & Lee, 2023; Gruver et al., 2023). One particularly compelling aspect of human cognition is the ability to self-correct. Self-correction (SC) is the continuous process of evaluating and refining one's responses or actions to achieve improved outcomes. There are generally two types of SC: *extrinsic* SC which utilizes the help of external knowledge and *intrinsic* SC which uses only one's own knowledge. Inspired by humans, the research community proposed a promising hypothesis that we could leverage the power of SC to enhance the quality of LLMs. In particular, researchers are interested in whether *intrinsic* SC ability could be applied in LLMs.

Over the last few years, numerous research studies have been conducted on the effectiveness of intrinsic SC of LLMs. For example, works like self-refine (Madaan et al., 2023), self-verification (Gero et al., 2023; Weng et al., 2023; Manakul et al., 2023), post-editing (Raunak et al., 2023), all demonstrate significantly enhanced performance of SC on respective benchmarks compared to results obtained without SC. However, recent works have questioned the efficacy of intrinsic SC mechanisms in LLMs (Huang et al., 2024; Kumar et al., 2024). A notable critique presented by (Huang et al., 2024) is that LLMs do not possess the ability to perform intrinsic self-correction. Their argument is based on a fundamental question: *why couldn't an LLM get the correct answer in the initial attempt instead of going through intrinsic SC?* They hypothesize that existing works on SC abilities in LLMs shall be attributed to three factors: external feedback (such as the existence of a retrieval model) that

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[1]whether LLMs exhibit emergent ability is still in debate in academia, partially due to the definition of "emergence". See Schaeffer et al. (2023) and Lu et al. (2023) for details.

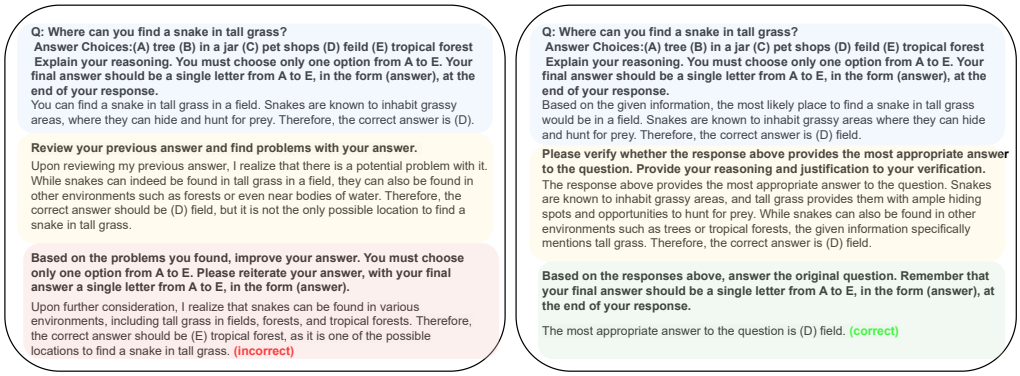

Figure 1: An example where the biased prompt (left) shifts the answer from correct to incorrect between the intrinsic SC stages, whereas our unbiased prompt (right) maintains the correct answer. The blue, yellow, and red regions correspond to Stage 1,2,3 in Section 2.2, respectively. Phrases such as "find problems" and "improve" might hint at an incorrect initial answer to the LLM and force it to change answers. On the other hand, the unbiased prompt avoids those unnecessary changes.

converts the problem to an extrinsic setting, ground-truth label as an early stopping criterion that hinders the practical usage, and uninformative prompts that limit LLM's ability when it generates the first response before SC (which raises the possibility of unfair comparison). Through the experiments, both Huang et al. (2024) and Kumar et al. (2024) report that LLMs showcase different levels of degradation in QA accuracy after intrinsic SC.

In light of such disagreement, it is important to answer a set of fundamental questions: **Does intrinsic SC ability exist within LLMs? And if it does, why couldn't an LLM get the correct answer in the initial attempt instead of going through intrinsic SC?**

This paper answers the above fundamental questions. With theoretical analyses and quantitative experiments under the same settings as Huang et al. (2024), we empirically show that LLMs **do** have intrinsic SC ability. In addition, we also show that LLMs cannot answer questions correctly during the initial attempt because of their inherent hallucination property.

Building on top of the answers, further investigations are conducted on the conditions that maximize SC potential within LLMs. Through theoretical analyses and empirical experiments, we identify two factors impacting the intrinsic SC ability of LLMs. The first factor is temperature. A non-zero temperature, such as done by Huang et al. (2024), degrades SC ability. The second factor, equally important, is the neutrality and fairness of the prompts used during various stages of SC. A fair and unbiased prompt that doesn't directly or indirectly influence the LLM to change or maintain its initial answer provides the most stable benefits from the intrinsic SC ability of LLMs. In Fig. (3), we provide an example illustrating how biased (Huang et al., 2024) and unbiased (ours) prompts affect the output of the LLM during different stages of SC.

This paper's contributions could be summarized as the following:

- We provide a novel perspective on the functioning nature of intrinsic self-correction in Section 2. We posit that self-correction abilities bear similarities to chain-of-thought (CoT) and self-verification prompting methodologies, which furnish LLMs with additional context for generating the final response.
- We highlight the importance of the temperature of LLMs in the success of intrinsic self-correction in Section 4. While increasing the temperature will benefit the creativity of response, it may be detrimental for SC, depending on the structure of LLM's response.
- Additionally, in Section 5, we show that the SC prompt needs to be unbiased so that LLMs do not accidentally attend to tokens that might trigger unwanted answer updates during the SC process. We also provide guidelines to compose unbiased prompts for future research.
- We empirically verify the effects of the two factors above, showcasing the accuracy degradation of generated SC responses when increasing temperature and utilizing biased prompts, and intrinsic SC can be achieved by LLMs via fair prompts and zero temperature.

Our answers to those fundamental questions serve as a basis for the underlying mechanism of intrinsic SC, and we hope that our findings will shed further light on LLM's abilities and encourage future works and discussions to explore methods that could leverage the existing SC ability more effectively.

2 New Perspective on Intrinsic Self Correction

2.1 Preliminary

The text generation process of LLMs could be modeled as an auto-regressive probabilistic distribution P. Given an LLM and a prompt of t tokens $\tau = \{w_1, w_2, ...w_t\}$, the auto-regressive generation of the response takes the form of Eq. (1), where y is the (unnormalized) logits from the LLM, w_{t+1} is a potential next token, $l_{w_{t+1}}$ is the scaled logit value for w_{t+1}, and T is the temperature. Thus, the generated content of an LLM adheres to the distribution of its logits, readjusted by the temperature. As temperature approaches 0, it is clear that the generation process changes from sampling in a distribution to selecting the index of the maximum value of logits, as described in Eq. (2).

$$P(w_{t+1}|w_1, ...w_t) \sim y \quad , \quad l_{w_{t+1}} = \frac{e^{\frac{y_{w_{t+1}}}{T}}}{\sum_{k=1}^{N} e^{\frac{y_k}{T}}} \tag{1}$$

$$\lim_{T \to 0^+} l_{w_{t+1}} = \begin{cases} 1 & \text{if } y_{w_{t+1}} = \max y \\ 0 & \text{otherwise} \end{cases} \tag{2}$$

Although the definitions above may appear elementary, they constitute critical foundations for this study. In particular, they provide justifications for some of the analyses in the future sections. We draw several insights on this foundational formation:

(1) Under $T = 0$, any arbitrary LLM should output the same response on some fixed prompt τ. Unfortunately, models of the GPT family (OpenAI, 2024; Brown et al., 2020) do not follow this principle, possibly due to their query optimizations and internal sparse mixture-of-experts (MoE) design of GPT-4. In particular, GPT-4 exhibits very high variance for its answers under $T = 0$. Those unknown behaviors cause some variations in the evaluation process.

(2) Intrinsic SC increases the prompt length t constructively, which helps improve the generation quality of the final response. This is similar to self-verification (Weng et al., 2023; Gero et al., 2023) and chain-of-thought (CoT) (Wei et al., 2022b). We will elaborate on this in Section 2.2.

(3) Increasing temperature might not cause visible accuracy degradation in the evaluation process for response generation without SC, but it might be detrimental to the intrinsic SC due to the non-optimal sampling process. We will elaborate on this in Section 4.

2.2 Does intrinsic SC exist in LLMs?

In this section, we answer the first fundamental question of whether intrinsic SC exists in LLMs by providing an alternative explanation of the underlying mechanism of self-correction. For now, we focus on one model (GPT3.5-0613) and fix the temperature at zero.

We initiate our analysis with the chain-of-thought (CoT) prompting technique [2], which is proven to be useful in enhancing LLM's performance in accuracy (Wei et al., 2022b). Based on the auto-regressive nature of LLM generation, Chu et al. (2023) models the CoT process as a conditional probability shown in Eq. (3), where A is the answer to the prompt (and question) τ_1, and R_1 is the rationale provided by LLM in the CoT process. The reasoning context introduced by CoT helps the LLM make a better final decision. Similarly, using the effective depth of the transformer circuit, Feng et al. (2023) argues that CoT increases the effective depth of the circuit by letting the generated outputs repeatedly loop back to the input [3].

[2] CoT prompt could be elicited with phrases such as "explain your reasoning" or "think step by step" in LLMs during the prompting.

[3] It shall be noted that Feng et al. (2023) actually proves that with longer generations, transformers (for **some** choice of weights; not necessarily the pre-trained weights) have more expressive capacity to model more complex problems.